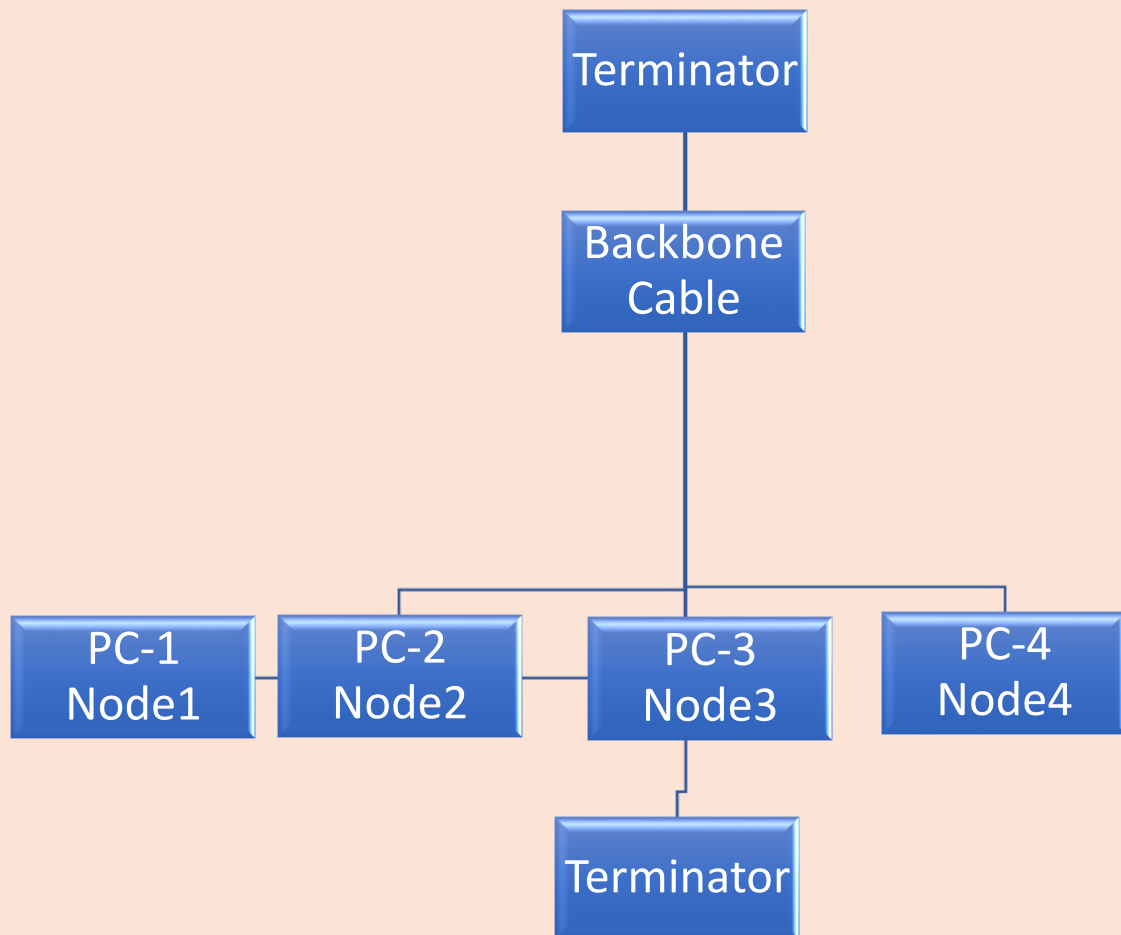
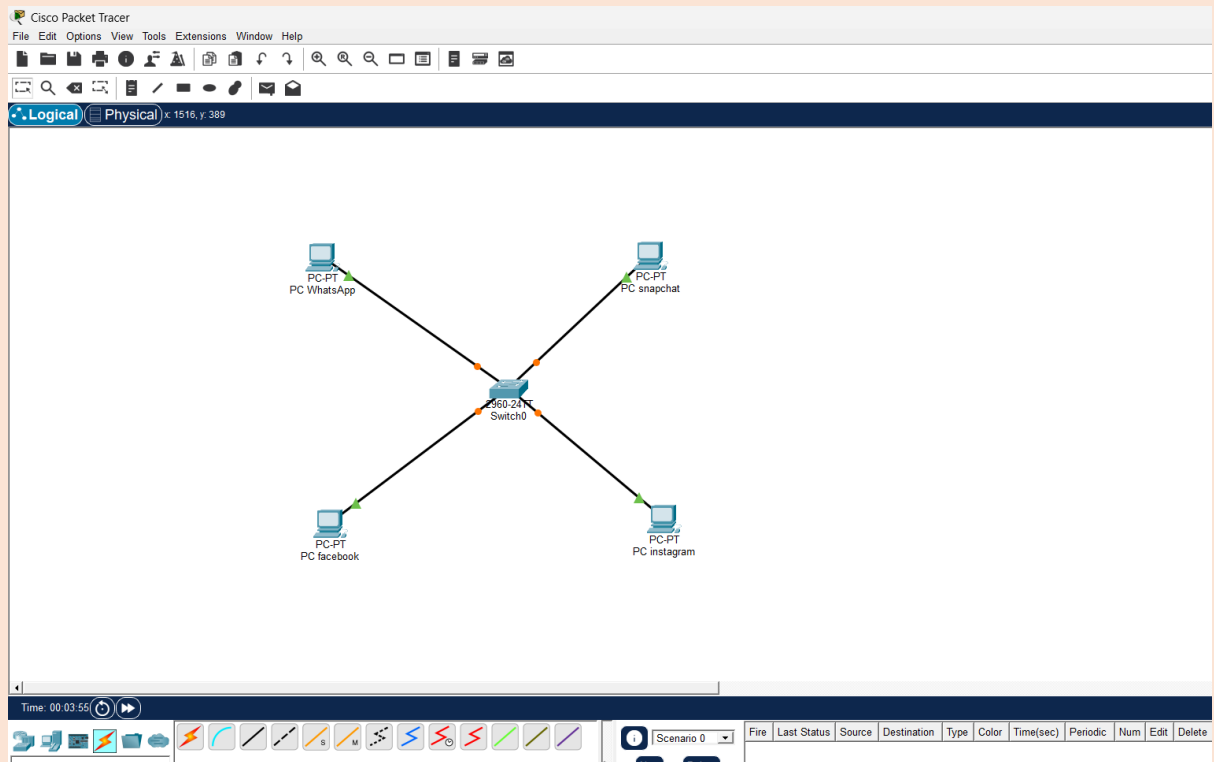


NETWORKING TOPOLOGIES THEORY

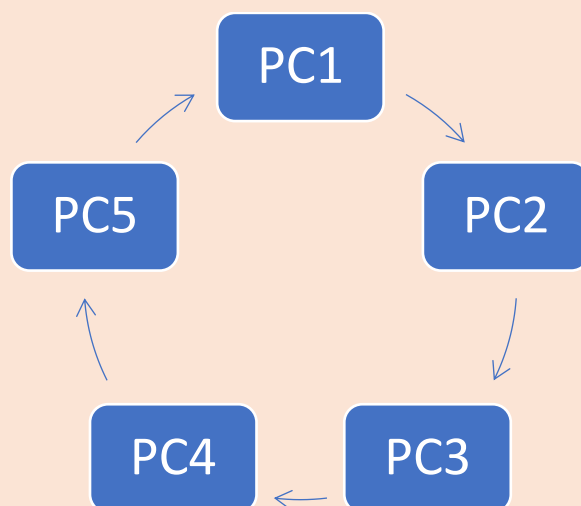
1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.



2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.



3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.
Hint: Use arrows to show the direction of data flow.



Data Flow:

PC-1 → PC-2 → PC-3 → PC-4 → PC-5 → PC-1

Explanation:

In a ring topology, each network device is connected to exactly two neighboring devices, forming a closed-loop communication path. Data packets are transmitted sequentially from one node to the next in a single direction until they reach the destination node. Each intermediate node receives, processes, and forwards the data to maintain continuous network communication.

4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format

Topology	Advantages	Disadvantages
Mesh Topology	1. High reliability because multiple paths are available for data transmission. 2. Failure of one connection does not affect communication between other devices.	1. Requires a large number of cables and ports, making installation expensive. 2. Complex to configure, manage, and maintain.
Star Topology	1. Easy to install, manage, and troubleshoot. 2. Failure of one device or cable does not affect the rest of the network.	1. If the central switch or hub fails, the entire network becomes unavailable. 2. Requires more cabling than a bus topology, increasing installation cost.